Belief-Augmented Reinforcement Learning for Pick-and-Place Under Observation Noise, Occlusion, and Target Drift

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***Abstract*—Robotic pick-and-place under partial observability is challenging because sensor noise, occlusion, and environmental disturbance corrupt or eliminate target state information, turning a standard MDP into a POMDP. This work formalizes the problem on a SO-ARM101 6-DOF arm in a physics-based MuJoCo environment with three simultaneous uncertainty sources: distance-dependent wrist-camera noise, occlusion by two distractor blocks, and stochastic target drift ($\sigma_{\text{drift}} = 2$ mm/step). Two learned policies are compared: plain PPO, which receives the raw (or stale) wrist observation directly, and belief-augmented PPO, which routes observations through a 300-particle bootstrap filter and exposes the filter mean $\boldsymbol{\mu}$ and standard deviation $\boldsymbol{\sigma}$ to the policy. A noise sweep across $\sigma_{\text{ep}} \in \{3, \ldots, 30\}$ mm reveals an unexpected result: plain PPO achieves a flat $\sim 82\%$ success rate independent of noise level, indicating it learned a proximity-sweep strategy that bypasses observation quality entirely. Belief PPO achieves its largest advantage of $+11\%$ at $\sigma_{\text{ep}} = 10$ mm, where the particle filter converges reliably, but degrades to -16% at $\sigma_{\text{ep}} = 30$ mm as the filter operates outside its training distribution. These results characterize the conditions under which belief augmentation provides benefit, and expose a fundamental evaluation limitation: proximity-based auto-grasp prevents the localization bottleneck from manifesting at the task-success level.**

I. Introduction

Robotic manipulation in unstructured environments requires reliable action despite imperfect sensing. Real wrist cameras return noisy pose estimates, distractor objects can occlude the target, and physical perturbations shift objects between observations. Treating such settings as fully observable MDPs ignores fundamental epistemic uncertainty and leads to brittle policies that degrade when sensor quality changes or objects move unexpectedly.

This work studies a concrete instantiation: pick-and-place of a target LEGO block in the presence of two occluding distractor blocks and stochastic target drift, using only a wrist-mounted camera whose noise scales linearly with end-effector-to-target distance. Three uncertainty sources are active simultaneously. *Distance-dependent sensor noise:* the camera returns a Gaussian-corrupted block pose; approaching improves observations, creating an implicit information-gathering incentive. *Occlusion:* when a distractor interposes between camera and target, the observation drops out entirely; the agent must reason from its last known estimate. *Target drift:* the block undergoes random XY perturbations ($\sigma_{\text{drift}} = 2$ mm/step) while not held, so even a recent accurate estimate decays as time passes.

Together these create a genuine exploration–exploitation tension in the POMDP sense: at each step the agent chooses between spending more steps refining its belief (approach for a better reading, reposition to clear occlusion) and committing to a grasp on its current, imperfect estimate. This mirrors the bandit arm-selection problem: each "pull" (approach step) reduces the uncertainty about the block's position, but each step also consumes horizon budget and risks triggering an incorrect auto-grasp. The optimal policy must balance these competing pressures, a decision that depends on the current belief uncertainty $\boldsymbol{\sigma}$ and how much remaining budget is available.

Two learned policies are compared. *Plain PPO* [4] treats the POMDP as a memoryless MDP, receiving the last visible wrist observation replayed during occlusion. It has no explicit mechanism to reason about uncertainty. *Belief-augmented PPO* maintains a bootstrap particle filter [3] over the target pose and feeds both the filter mean and its uncertainty estimate directly into the policy network, enabling explicit uncertainty-conditioned behavior without an external memory or recurrent architecture.

The main finding is that plain PPO learned to bypass the observation entirely via a proximity-sweep strategy, achieving noise-invariant performance across all tested levels. Belief PPO is genuinely observation-dependent, outperforms at moderate noise ($+11\%$ at $\sigma_{\text{ep}} = 10$ mm), but degrades at noise levels beyond its training distribution where the particle filter cannot converge.

Departures from the proposal. The original proposal specified two uncertainty sources; a third (target drift) was added to increase task difficulty and validate belief tracking under nonstationary targets, which is more representative of real-world scenes. The end-effector Cartesian position was removed from the observation vector after identifying it as redundant with joint angles via forward kinematics; this reduced dimensionality from 16D/19D to 13D/16D. The expected finding—that belief PPO uniformly outperforms—did not hold; the actual noise-regime-dependent result is the central empirical contribution.

II. Background and Related Work

POMDPs for robotic manipulation. The POMDP framework [1] provides the formal foundation for sequential decision-making under state uncertainty. Kaelbling et al. define the belief-state MDP reduction in which the agent maintains a posterior over hidden states and acts on that distribution, guaranteeing that a sufficiently expressive belief representation yields an optimal policy. In robotic manipulation, partial observability arises from sensor noise, occlusion, and contact uncertainty. Koval et al. [2] model contact-induced uncertainty as belief-space transitions, enabling planning over pose distributions rather than point estimates. This work addresses the complementary case of visual uncertainty: wrist-camera noise and occlusion in a non-contact approach phase.

Belief-state representations. The bootstrap particle filter [3] approximates the belief as a weighted set of samples and naturally handles arbitrary non-Gaussian distributions, including the multimodal beliefs that arise when a target has been occluded long enough to occupy any location in a region. Kalman-family filters (KF, EKF, UKF) are computationally cheaper but require unimodal Gaussian beliefs—an assumption that breaks after extended occlusion. In this environment, the two distractor blocks can occlude the target for 50+ consecutive steps, making the particle filter the appropriate representation.

Deep RL for manipulation. Proximal policy optimization [4] is widely used for continuous-action robotic control due to its stability and relative ease of hyperparameter tuning. Reward shaping is known to be both critical and dangerous: Ng et al. [5] prove that only potential-based shaping preserves the set of optimal policies, while arbitrary dense rewards can introduce spurious local optima that trap the agent. This guarantee was the design basis for the approach and carry shaping terms used here; five distinct reward-hacking exploits discovered during development were each resolved by converting the offending term to a potential-based or one-time milestone form.

Online POMDP planning. Silver and Veness [6] introduced POMCP, extending Monte Carlo tree search to POMDPs by using particle filters as belief representations within the tree. Sunberg and Kochenderfer [7] extend this to continuous observation spaces via double progressive widening (POMCPOW), which prevents the search tree from blowing up on every distinct observation. POMCP and POMCPOW provide the theoretical upper bound for what an online planner with direct simulator access could achieve on this task.

III. Problem Formulation

A. POMDP Definition

The task is formalized as a POMDP tuple $(\mathcal{S}, \mathcal{A}, \mathcal{O}, T, Z, R, \gamma)$.

State space $\mathcal{S}$. The full (partially hidden) system state is

$$s = \big(\mathbf{q},\ \mathbf{p}_{\text{blk}},\ \mathbf{p}_{d_1},\ \mathbf{p}_{d_2},\ h\big), \tag{1}$$

where $\mathbf{q} = [q_1, \ldots, q_5, g] \in \mathbb{R}^6$ contains the five arm joint angles and the gripper state $g \in \{0,1\}$ (both *fully observed* via joint encoders); $\mathbf{p}_{\text{blk}} = (x_b, y_b, \theta_b) \in \mathbb{R}^3$ is the target block pose in the world frame (*hidden*); $\mathbf{p}_{d_1}, \mathbf{p}_{d_2} \in \mathbb{R}^3$ are the distractor block poses (*hidden*); and $h \in \{0,1\}$ is a binary flag indicating whether the target is currently gripped (*observed*: determined by proximity, not a sensor reading).

Action space $\mathcal{A}$. Actions are continuous 4-vectors

$$a = \big[\Delta x,\ \Delta y,\ \Delta z,\ g_{\text{cmd}}\big] \in \mathbb{R}^4, \tag{2}$$

where $\Delta x, \Delta y, \Delta z \in [-0.02, 0.02]$ m are Cartesian end-effector deltas and $g_{\text{cmd}} \in [-1, 1]$ is the gripper command. The environment resolves joint angles internally via geometric inverse kinematics; the policy operates entirely in Cartesian space and never directly commands joints. Grasping is proximity-triggered: when the EE enters a sphere of radius $r_{\text{grasp}} = 15$ mm around any block, that block is automatically attached via a rigid position constraint. The policy therefore controls approach and positioning, not the grasp trigger directly.

Observation space $\mathcal{O}$. The agent receives a partial observation

$$o_t = \begin{cases} \big[\underbrace{\mathbf{q}}_{6},\ \underbrace{\hat{\mathbf{p}_{\text{blk}}}^{xy\theta}}_{3},\ \underbrace{\mathbf{p}_{\text{goal}}}_{2},\ h,\ c\big] & \text{plain PPO,} \\ \big[\underbrace{\mathbf{q}}_{6},\ \underbrace{\boldsymbol{\mu}}_{3},\ \underbrace{\boldsymbol{\sigma}}_{3},\ \underbrace{\mathbf{p}_{\text{goal}}}_{2},\ h,\ c\big] & \text{belief PPO,} \end{cases} \tag{3}$$

giving $o_t \in \mathbb{R}^{13}$ or $\mathbb{R}^{16}$ respectively. Here $\hat{\mathbf{p}_{\text{blk}}}$ is the (stale/noisy) wrist-camera reading of the target pose, or the particle filter mean $\boldsymbol{\mu}$ in belief mode; $\boldsymbol{\sigma} \in \mathbb{R}^3$ is the per-dimension particle filter standard deviation (belief mode only); $\mathbf{p}_{\text{goal}} \in \mathbb{R}^2$ is the fixed goal position in the world frame; and $c \in \{0,1\}$ is a binary flag that is 1 when the wrist camera is occluded. The EE Cartesian position is intentionally omitted: it is a deterministic function of $\mathbf{q}$ via forward kinematics and carries no additional information.

Transition model T. State transitions are defined implicitly by the MuJoCo physics engine (`mj_step`, 10 substeps at $\Delta t = 0.0005$ s each). Robot joint angles update deterministically from the IK-resolved target. When not held, the target block undergoes stochastic drift:

$$\mathbf{p}^{xy}_{\text{blk},t+1} = \mathbf{p}^{xy}_{\text{blk},t} + \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0},\ \sigma^2_{\text{drift}}\mathbf{I}_2), \tag{4}$$

with $\sigma_{\text{drift}} = 2$ mm/step. Distractor blocks remain stationary throughout the episode.

Observation function Z. The wrist-camera model is

$$P(o_t \mid s_t) = \begin{cases} \mathcal{N}\big(\mathbf{p}^{xy\theta}_{\text{blk}},\ \Sigma_{\text{obs}}(d_t)\big) & \text{target visible,} \\ \delta(o_t - o_\tau) & \text{target occluded,} \end{cases} \tag{5}$$

where $\Sigma_{\text{obs}}(d_t) = \operatorname{diag}\big(\sigma^2_{\text{obs}}, \sigma^2_{\text{obs}}, (10\sigma_{\text{obs}})^2\big)$, $\sigma_{\text{obs}}(d_t) = \max\big(\sigma_{\text{ep}},\ 0.008 \cdot (d_t/0.15)\big)$, d_t is the current EE-to-target distance, and o_τ is the last visible reading replayed during occlusion. The per-episode noise floor $\sigma_{\text{ep}} \sim \mathcal{U}(3, 15)$ mm is drawn once at episode start and held fixed throughout. Occlusion is determined geometrically: a frustum-based virtual wrist camera with 25° downward pitch and target-tracking yaw checks whether any distractor's footprint projected onto the image plane overlaps the target's footprint.

Reward R and discount γ. The reward function is detailed in Table I. The discount factor is $\gamma = 0.99$ with a maximum episode horizon of $T = 300$ steps, after which the episode terminates as a timeout failure.

B. Three Sources of Partial Observability

Three distinct uncertainty sources make this task a genuine POMDP and together produce the explore–exploit tension central to this work.

1. Distance-dependent sensor noise. The effective observation standard deviation $\sigma_{\text{obs}}(d_t)$ grows with distance, so the agent can actively reduce its own measurement uncertainty by moving closer. This creates a bandit-like incentive: approach more to localize better before grasping (exploration), or commit to the current noisy estimate (exploitation). The optimal exploration depth is a function of the current belief spread $\boldsymbol{\sigma}$, the inter-block spacing, and the remaining horizon.

2. Occlusion-induced observation dropout. When a distractor interposes between the camera and the target, the observation is completely absent for that step. The agent cannot resolve this by approaching; it must reposition to clear the distractor from the camera's line of sight. Unlike sensor noise, occlusion uncertainty is *not reducible by approaching the target*—it requires a lateral repositioning maneuver that may move the EE away from the target, trading off placement progress for better information. Per-step occlusion rates of 10–25% are observed under trained policies.

3. Stochastic target drift. The target block drifts randomly by $\sigma_{\text{drift}} = 2$ mm/step while ungripped, making localization a moving-target problem. A confident belief based on a reading taken T_{occ} steps ago has degraded by approximately $\sqrt{T_{\text{occ}}} \cdot \sigma_{\text{drift}}$ in expected position error. This third source was added over the original proposal to validate that the particle filter's predict step can track a genuinely nonstationary target; without drift, a simple last-observation cache is close to optimal.

C. Reward Function

The approach shaping (rows 2–5) follows the potential-based form of Ng et al. [5]: rewarding the improvement in a scalar potential rather than the absolute value ensures that lingering near the block provides no benefit beyond the one-time reach milestone. Without this design, early experiments found the agent farming per-step proximity rewards by hovering above the block without grasping. The carry phase (rows 8–9) uses the same pattern once the target is held. Terminal rewards (rows 10–13) are large relative to shaping terms to ensure the episodic objective dominates.

D. Simulation Environment

The SO-ARM101 is a 5-DOF desktop arm with a parallel-jaw gripper (Fig. 1). The MuJoCo scene contains a red 2×4 LEGO brick (target, 32×16 mm footprint, 11 mm tall), two blue 2×2 LEGO bricks (distractors, 40×20 mm footprint, 33 mm tall for exaggerated wrist-camera occlusion), and a fixed yellow goal marker. At each episode reset, the target is placed uniformly within the arm's cylindrical reachable workspace ($r \in [12, 22]$ cm, ± 1 rad from front); distractors spawn uniformly within 90 mm of the target with a minimum 50 mm inter-block separation. The goal is sampled at least 100 mm from any block. The virtual wrist camera is co-located with the EE, pointed at the target with a fixed 25° downward pitch; its yaw tracks the target so that the target is always inside the frustum when not occluded.

TABLE I
REWARD FUNCTION. SHAPING ROWS MARKED ($\star$) USE POTENTIAL-BASED FORM $w \cdot \text{clip}(\Delta d/20\text{MM}, -1, 1)$; BONUS ROWS MARKED (†) FIRE ONCE PER EPISODE ON FIRST ENTRY. d_{xy}: EE-TO-BLOCK XY DISTANCE. d_g: EE-TO-GOAL DISTANCE.

Component	Value	Trigger
Step cost	-1.0	Every step
XY approach*	± 3.0	Per step, pre-grasp
Proximity bonus	$\leq +2.0$	$d_{xy} < 25$ mm
Close-in bonus	$\leq +3.0$	$d_{xy} < 12$ mm
Reach block†	$+5.0$	$d_{xy} < 15$ mm
Grasp target	$+10.0$	Correct block
Grasp distractor	-15.0	Wrong block (terminates)
Carry shaping*	± 5.0	Per step, while holding
Near goal†	$+5.0$	$d_g < 30$ mm, holding
Perfect placement	$+50.0$	Release, $d_g < 10$ mm
Precise placement	$+30.0$	Release, $d_g < 20$ mm
Close placement	$+10.0$	Release, $d_g < 40$ mm
Missed release	-10.0	Release, $d_g \geq 40$ mm

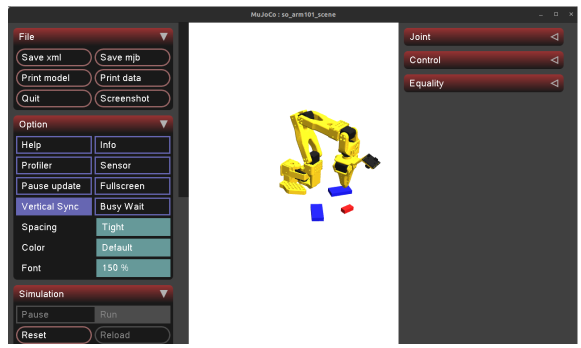


Fig. 1. MuJoCo simulation environment. The SO-ARM101 arm must grasp the red 2×4 LEGO target block and carry it to the yellow goal marker. Two taller blue 2×2 LEGO bricks serve as distractor/occluder objects, spawned within 90 mm of the target at each episode reset.

IV. Solution Approach

A. Plain PPO Baseline

The plain PPO agent treats the POMDP as a memoryless MDP: during occlusion it receives the cached last-visible wrist observation o_τ as if it were the current reading. The 13-dimensional observation is given by (3) (plain case). The policy is a two-layer MLP with 256 hidden units per layer and ReLU activations, matching the SB3 default architecture.

Training uses Stable-Baselines3 [8] PPO with $n_{\text{env}} = 8$ parallel environments (SubprocVecEnv), $n_{\text{steps}} = 4096$ per rollout, mini-batch size 256, $n_{\text{epochs}} = 10$, entropy coefficient

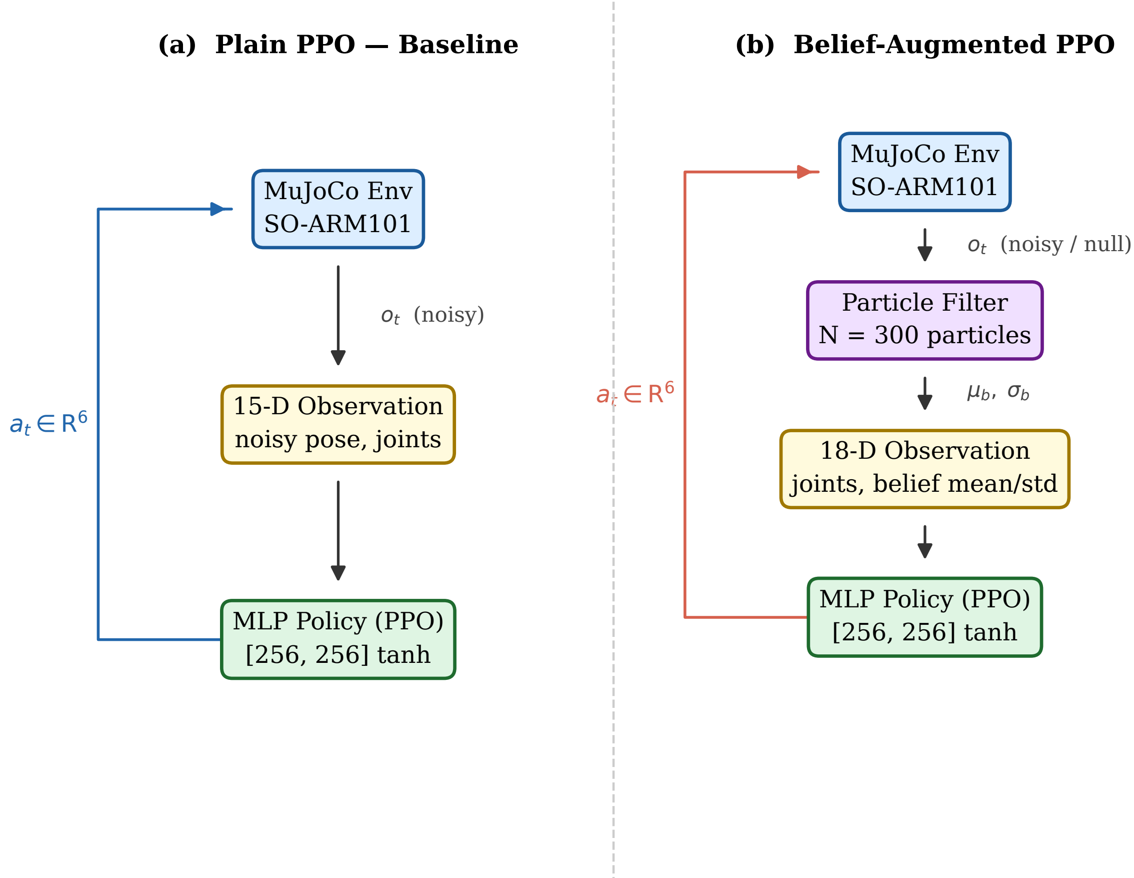


Fig. 2. Architecture comparison. (a) Plain PPO exposes the noisy or stale wrist-camera reading as a 13-D vector directly to the MLP policy. (b) Belief-augmented PPO routes each step's observation through a 300-particle bootstrap filter; the filter summary $(\boldsymbol{\mu}, \boldsymbol{\sigma})$ augments the 16-D policy input, giving the network an explicit uncertainty signal it can condition on.

0.02, $\gamma = 0.99$, clip ratio 0.2, linearly-decayed learning rate $3 \times 10^{-4} \to 0$, and VecNormalize applied to both observations and rewards (clip threshold ± 10). Training runs for 5M environment steps; an EvalCallback saves the best checkpoint based on mean reward over 50 evaluation episodes every 10k training steps.

B. Belief-Augmented PPO

1) Bootstrap Particle Filter: A bootstrap (SIR) particle filter [3] maintains a belief over the target's hidden pose $\mathbf{p}_{\text{blk}} = (x_b, y_b, \theta_b)$ using $N = 300$ weighted particles $\{x^{(i)}, w^{(i)}\}_{i=1}^{N}$. The filter runs synchronously at each environment step.

Predict. Each particle is perturbed to account for target drift and model uncertainty:

$$x_{t|t-1}^{(i)} = x_{t-1}^{(i)} + \epsilon, \quad \epsilon \sim \mathcal{N}\big(\mathbf{0},\, \text{diag}(\sigma_p^2, \sigma_p^2, \sigma_{p\theta}^2)\big), \tag{6}$$

with $\sigma_p = \max(0.5 \text{ mm}, \sigma_{\text{drift}}) = 2$ mm and $\sigma_{p\theta} = 0.01$ rad. Setting $\sigma_p \geq \sigma_{\text{drift}}$ ensures the filter's spread does not lag behind the actual block drift.

Update. When the target is visible ($c_t = 0$), particle weights are updated by the Gaussian likelihood:

$$w^{(i)} \propto w^{(i)} \cdot \mathcal{N}\big(o_t^{xy};\, x_{xy}^{(i)},\, \sigma_{\text{lik}}^2 \mathbf{I}\big), \quad \sigma_{\text{lik}} = 1.5\, \sigma_{\text{obs}}(d_t). \tag{7}$$

The $1.5\times$ likelihood widening prevents particle impoverishment: using the exact sensor sigma collapses the effective sample size $N_{\text{eff}} = 1/\sum_i (w^{(i)})^2$ to near 1 each step, degenerating the filter into a noisy running average. During occlusion ($c_t = 1$), the update step is skipped entirely; particles spread via the predict step alone, correctly growing $\boldsymbol{\sigma}$ without incorporating spurious information.

Resample and reinject. Systematic resampling is applied when $N_{\text{eff}} < N/2$. After resampling, 5% of particles are re-injected near the last visible observation to prevent degeneracy during extended occlusion.

Belief summary. The filter output is compressed to $(\boldsymbol{\mu}, \boldsymbol{\sigma}) \in \mathbb{R}^3 \times \mathbb{R}^3$:

$$\mu_k = \sum_i w^{(i)} x_k^{(i)}, \quad \sigma_k = \sqrt{\sum_i w^{(i)} \big(x_k^{(i)} - \mu_k\big)^2}, \tag{8}$$

for each dimension $k \in \{x, y, \theta\}$. Large $\boldsymbol{\sigma}$ indicates recent or extended occlusion, or accumulated drift uncertainty; small $\boldsymbol{\sigma}$ indicates confident localization.

2) Policy Architecture and Training: The 16-dimensional belief observation is given by (3) (belief case). The critical addition over plain PPO is $\boldsymbol{\sigma}$: the policy can learn to act

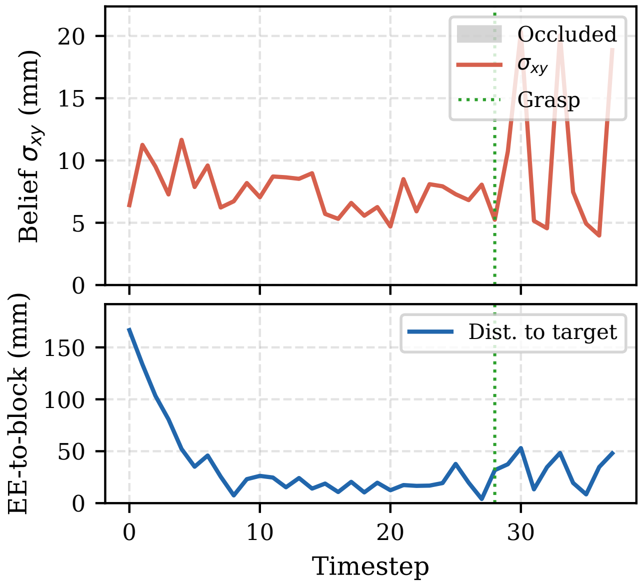
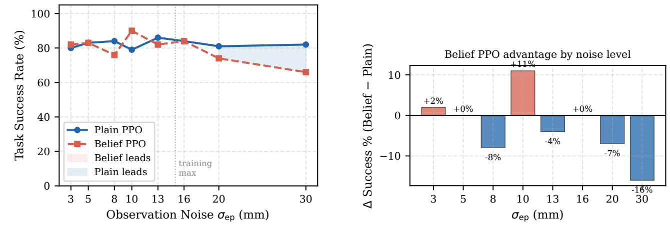


Fig. 3. Belief σ_{xy} (top) and EE-to-block distance (bottom) over a representative successful episode. Gray shaded spans mark wrist-camera occlusion windows. Uncertainty grows during each occlusion as particles spread via the predict step alone. Once occlusion clears and the agent approaches, σ_{xy} collapses rapidly as the distance-dependent noise model produces tight likelihoods at close range. The dashed green line marks the auto-grasp trigger.

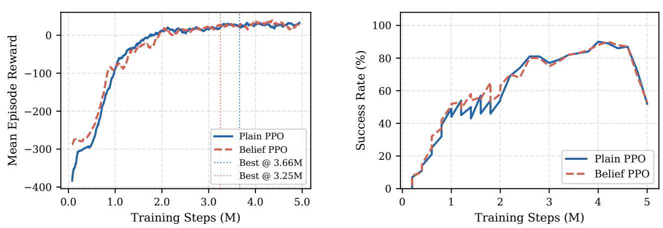


(a) Success rate vs. σ_{ep} (b) Belief − Plain advantage

Fig. 5. Noise sweep results over 100 episodes per level (seed 42). Left: absolute success rates; the vertical dashed line marks the training maximum ($\sigma_{ep} = 15$ mm). Right: per-level advantage of belief over plain PPO.

TABLE II
NOISE SWEEP RESULTS (100 EPISODES, SEED 42). **BOLD** = HIGHER SUCCESS RATE. Δ = BELIEF − PLAIN.

σ_{ep} (mm)	Success (%)↑			Mean ep. len.	
	Plain	Belief	Δ	Plain	Belief
3	**80**	82	+2	76.2	69.9
5	83	**83**	0	65.3	70.2
8	**84**	76	−8	65.2	85.0
10	79	**90**	**+11**	75.8	45.5
13	**86**	82	−4	57.0	72.4
16	**84**	**84**	0	66.1	63.5
20	**81**	74	−7	72.4	93.3
30	**82**	66	−16	68.3	116.4



(a) Mean episode reward (b) Task success rate

Fig. 4. Training curves (15-step moving average). Dotted vertical lines mark the saved best-model checkpoints: plain PPO at 3.66M steps (best reward 68.3) and belief PPO at 3.25M steps (best reward 71.7).

cautiously when uncertainty is high (slowing the approach to avoid a wrong auto-grasp) and decisively when the filter is confident. No explicit information-gathering reward is used; this uncertainty-aware behavior must emerge from the task reward alone.

Reward shaping terms in belief PPO use the PF mean $\boldsymbol{\mu}$ in place of the raw sensor reading when computing the approach distance gradient. This provides a lower-variance reward signal at high noise levels, where the raw observation can mislead shaping by several centimeters. All PPO hyperparameters are identical to plain PPO; the network architecture is the same MLP $[256, 256]$. The agent trains for 5M steps.

V. RESULTS

A. Training Convergence

Training curves are shown in Fig. 4. Both agents discover the approach-shaping gradient around 1M steps, after which mean reward rises toward positive values. Plain PPO reaches a peak mean reward of 68.3 at step 3.66M; belief PPO achieves a slightly higher peak of 71.7 at step 3.25M, reflecting that the particle filter provides a cleaner reward gradient through filtered shaping. All evaluations below use the saved best-model checkpoint rather than the final checkpoint, which exhibits modest regression as the linear learning rate schedule approaches zero. Training variance is high throughout (±130 reward units) due to the bimodal return distribution: successful episodes yield +60–+100, while wrong-block grasps terminate early at approximately −250.

B. Noise Sweep: Main Result

The best-model checkpoints are evaluated at eight noise levels with 100 episodes each. Results are in Fig. 5 and Table II.

The most striking finding is that plain PPO achieves near-constant success (79–86%) across all eight noise levels, including $\sigma_{ep} = 30$ mm, which is twice the training maximum. This noise-invariant behavior indicates the policy learned a *proximity-sweep strategy*: the potential-based approach reward guides the EE into the block's approximate XY region, and the 15 mm auto-grasp threshold is large enough to trigger once the EE reaches that region, regardless of how corrupted the observation was. Accurate localization is simply not required when the grasp threshold exceeds the typical residual position error under this reward structure.

Belief PPO exhibits the expected observation-dependent behavior. At $\sigma_{\text{ep}} = 10$ mm, the particle filter converges to a low-error estimate within a few visible steps, enabling a decisive approach (mean 45.5 steps vs. 75.8 for plain PPO) and the largest advantage of $+11\%$. At $\sigma_{\text{ep}} = 30$ mm, the filter operates far outside its training distribution: the likelihood at $\sigma_{\text{lik}} = 45$ mm is nearly flat, particle weights barely update, and the cloud cannot concentrate. The policy responds by extending its search (mean 116.4 steps vs. 68.3 for plain PPO), and success falls to 66% vs. 82%. Within the training range ($\sigma_{\text{ep}} \leq 15$ mm), the two methods are roughly comparable, with belief PPO winning or tying at five of the six in-distribution noise levels.

Zero wrong-block grasps are recorded for either policy at any noise level; the grasp rate and task success rate are identical to within measurement noise throughout Table II. This confirms that the performance bottleneck is entirely in the *pre-grasp navigation* phase: once any agent reaches the target block to within 15 mm, it always grasps the correct block and places it successfully. The discrimination and placement sub-tasks are effectively solved; all remaining failures are navigation timeouts.

C. Belief Tracking Behavior

Fig. 3 shows the particle filter in operation over a representative successful episode. During each occlusion window (gray shading), σ_{xy} grows steadily as the predict-only step spreads particles to account for accumulated drift uncertainty. When occlusion clears and the agent begins approaching, σ_{xy} collapses rapidly: at close range, $\sigma_{\text{obs}}(d_t)$ is small, so the Gaussian likelihood peaks sharply and quickly concentrates the particle cloud. The auto-grasp fires immediately after σ_{xy} reaches a low value, consistent with the policy having learned an implicit confidence threshold below which it commits to the grasp. This sequence—uncertainty growth during occlusion, rapid collapse on approach, then decisive commitment—is the qualitatively correct behavior for the explore–exploit trade-off in this POMDP.

VI. Discussion and Conclusion

The central finding of this work is that proximity-based auto-grasp fundamentally changes what task success measures. With a 15 mm grasp threshold and potential-based approach shaping, any policy that reaches the approximate block region will succeed, regardless of how accurately it localized the block. Plain PPO learned exactly this: a robust proximity-sweep that ignores observation quality and achieves a flat success curve from $\sigma_{\text{ep}} = 3$ mm to $\sigma_{\text{ep}} = 30$ mm. This is not a failure of plain PPO—it found the optimal strategy for the task as designed—but it reveals a benchmark design limitation: if the grasp threshold is large relative to the localization error, the POMDP structure does not show up in the task metric.

Belief PPO, by contrast, genuinely uses observations. The correlation between σ_{ep} and episode length in Table II (belief: 45.5 to 116.4 steps; plain: 57–76 steps with no trend) is behavioral evidence that the policy conditions its commit decision on $\boldsymbol{\sigma}$: at high noise it searches longer before committing, exactly as an uncertainty-aware agent should. The degradation at $\sigma_{\text{ep}} = 30$ mm is not a fundamental flaw in the belief-augmented approach but a distribution-shift failure: the particle filter was trained and tuned for $\sigma_{\text{ep}} \leq 15$ mm, and the likelihood widening factor ($1.5\times$) that prevents impoverishment in-distribution becomes insufficient at twice the training noise level. Extending the training noise range or using a noise-adaptive likelihood width would address this.

Reward engineering was the dominant development cost. Over 18 training runs, five classes of reward-hacking exploit were identified and resolved. In every case the exploit was enabled by a per-step reward for occupying a desired state (near the block, gripper closed near block, near goal while holding): the agent learned to enter the state and remain there. The fix was always either a one-time milestone (fires only on the first entry into the state per episode) or a potential-based shaping term (rewards only the improvement, not the absolute value) [5]. The former eliminates indefinite state occupancy; the latter provides dense gradient signal while preserving the optimal policy.

Several limitations remain. The grasp check is two-dimensional (XY proximity only), a pragmatic choice after multiple failed attempts with a full 3D check: with a 3D condition, the approach reward provides no gradient for the z descent phase, since the 3D proximity signal is zero when hovering directly above the block. Real-robot evaluation via the existing ROS2 integration was begun but not completed; sim-to-real gaps in the noise model and block drift dynamics remain untested. Future work should tighten the grasp threshold (e.g., 5 mm) to force accurate localization to be task-critical, which would expose the full benefit of belief augmentation at the task-success level. Curriculum-based noise annealing and a noise-adaptive likelihood width are straightforward extensions that would improve out-of-distribution belief PPO performance.

Contributions and Release

This was a solo project. The MuJoCo simulation environment was designed and built from scratch (`lego_pick_env.py`), including the three-phase reward function, the multi-block frustum occlusion model, proximity-triggered constraint grasping, stochastic target drift, and the full VecNormalize training pipeline. The bootstrap particle filter (`particle_filter.py`) was implemented independently, including the predict/update/systematic-resample cycle, particle reinjection, and the widened-likelihood impoverishment heuristic. The geometric occlusion checker (`occlusion.py`), the analytical geometric IK (`compute_workspace.py`), and the noise-sweep evaluation harness (`eval_noise_sweep.py`) were all implemented from scratch. The PPO algorithm itself is from

Stable-Baselines3 [8]; the MuJoCo physics engine and the SO-ARM101 robot URDF are third-party assets.

The authors grant permission for this report to be posted publicly.

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